

Chapter 1: Statistical Sampling and Data Collection

A Level Statistics Notes · Rexponential

Why this chapter matters

Before any calculation, you must be able to critique how data was collected — this is examined as bookwork (definitions, evaluating a sampling method) just as often as through calculation, so it deserves proper revision time rather than being treated as common sense.

The other important feature of the course is the **large data set**. Every UK exam board publishes a real data set (Edexcel's is anonymised daily weather data from several UK and overseas locations) that you study throughout the course and which the exam assumes familiarity with — questions can reference its variables, units, and known quirks (missing values coded as “n/a”, categorical wind direction, etc.) without re-explaining them, so it's worth getting to know well before the exam rather than leaving it as last-minute revision.

1.1 Populations, samples and sampling frames

Concept

A **population** is the entire set of items or people a study is about. A **census** surveys every member of the population; a **sample** surveys a subset chosen to represent it. In practice a census is rarely used — it is expensive, slow, and often destructive (you cannot test the lifespan of every light bulb a factory makes and have any left to sell) — so almost all real statistics relies on sampling.

A **sampling frame** is a list of all the members of the population from which a sample is drawn (e.g. a school's register, the electoral roll, a list of order numbers). A good sampling frame is complete, accurate and up to date; a poor one introduces **bias** — a systematic (not random) tendency for the sample to misrepresent the population, which no amount of extra sampling can fix.

Key vocabulary

Term	Meaning
Population	The entire group being studied
Census	Data collected from every member of the population
Sample	A subset of the population, used to make inferences about it
Sampling frame	A list identifying every member of the population that can be sampled
Bias	A systematic error that makes a sample unrepresentative of the population

Procedure

When a question asks you to evaluate a sampling situation:

1. State whether a census or a sample is more appropriate, and justify it using cost, time, or the practicality of testing (destructive testing almost always rules out a census).

2. Identify what the sampling frame would be for the given context — be specific (“the school's Year 12 register”, not just “a list of students”).
3. Check the sampling frame against the population it needs to represent: does it miss anyone, or include people who shouldn't be there? Any mismatch is a source of bias.

Worked example

EXAMPLE 1.1

A mobile phone company wants to find out how satisfied its customers are with its network coverage. It has a database of all customers who have called its helpline in the last month. (a) Suggest, with a reason, whether the company should use a census or a sample. (b) The company decides to use the helpline database as its sampling frame. Explain why this could lead to a biased sample.

Step 1 (part a). A sample is more appropriate: the company likely has a very large number of customers, so contacting every one would be costly and time-consuming, and a well-chosen sample can still give a reliable estimate of overall satisfaction. (B1)

Step 2 (part b). Customers who call the helpline are more likely to be those who have had a problem, so they are not representative of the whole customer population — satisfied customers who never call are excluded from the frame entirely. (B1 for identifying the frame excludes/over-represents a specific group, B1 for linking this to biased (not just “inaccurate”) results.)

Answer (a) Use a sample — a census would be impractical given the likely size of the customer base. (b) The helpline database over-represents dissatisfied customers and excludes customers who never contact the helpline, so the sample would be biased towards negative satisfaction scores.

EXAM TIP

This kind of “explain why this could be biased” question has no calculation at all, and it is easy to under-prepare for it because it looks too easy to be worth revising. Always name which group is over- or under-represented, rather than writing vaguely that the sample “might not be accurate” — mark schemes specifically reward identifying the direction of the bias.

1.2 Sampling techniques

Concept

Once a sampling frame exists, there are several standard methods for choosing which members of it are actually sampled. Each trades off cost, ease of implementation, and how well it protects against bias. The A Level spec expects you to describe how each is carried out, evaluate its suitability for a given context, and — for stratified sampling — carry out the proportional allocation calculation.

Key vocabulary

Method	How it works	Advantage	Disadvantage
Simple random sampling	Every member of the frame is numbered, then a random number generator or lottery method is used to choose the sample, so that each member is equally likely to be selected	Free from selection bias	Needs a complete sampling frame; can be impractical for large populations
Systematic sampling	Members are numbered and a fixed interval $k = (\text{population size}) \div (\text{sample size})$ is calculated, and every k -th member is then sampled, starting from a random point	Simple to carry out, spreads the sample evenly	Can introduce bias if the frame has a hidden periodic pattern matching the interval
Stratified sampling	The population is split into non-overlapping strata (groups) by a characteristic (e.g. year group); a simple random sample is then taken from each stratum, with the number taken proportional to that stratum's size	Ensures every subgroup is represented in proportion to the population	Requires the population to be clearly divided into strata beforehand
Quota sampling	An interviewer is given quotas (e.g. 20 men aged 18–30) to fill using non-random judgement, until each quota is met	Quick, cheap, no sampling frame needed	Non-random — introduces interviewer bias, cannot calculate reliable error margins
Opportunity (convenience) sampling	The sample consists of whoever is available and willing at the time and place of sampling	Easy and cheap to carry out	Highly likely to be unrepresentative and biased

Procedure (stratified sampling calculation)

1. Confirm the strata and their sizes, and find the total population size by summing the strata.
2. Find the sampling fraction: $(\text{required sample size}) \div (\text{total population size})$.
3. Multiply each stratum's size by the sampling fraction to get the number to sample from that stratum.

4. Round each result to the nearest whole number (state that you have rounded — this is a common source of lost marks if the rounded values don't sum exactly to the target sample size, which is acceptable and expected).
5. Within each stratum, select the required number using simple random sampling.

Key formula

KEY FORMULA

$$n_h = n \times \frac{N_h}{N}$$

where:

- n_h = the sample size taken from stratum h
- n = the total sample size
- N_h = the population size of stratum h
- N = the total population size

Worked example

EXAMPLE 1.2

A school has 180 students in Year 12, 165 in Year 13, and no other year groups take A Level. A researcher wants a stratified sample of 50 students, stratified by year group. Calculate the number of students that should be sampled from each year group.

Step 1. Total population = $180 + 165 = 345$. (M1)

Step 2. Calculate the sampling fraction. (M1)

$$\text{Sampling fraction} = \frac{50}{345}$$

Step 3. Allocate the sample to each stratum. (A1 A1)

$$n_{Y12} = 180 \times \frac{50}{345} = 26.09... \rightarrow 26$$

$$n_{Y13} = 165 \times \frac{50}{345} = 23.91... \rightarrow 24$$

Answer 26 students from Year 12 and 24 students from Year 13 ($26 + 24 = 50$, as required).

EXAM TIP

It is easy to be confident with the arithmetic here but lose marks on the describe/evaluate questions by memorising generic sentences about each method without understanding *why* each disadvantage arises. Justify each disadvantage from first principles (e.g. “why does quota sampling introduce bias?”) rather than reciting a learned phrase — examiners reward reasoning that fits the specific scenario in the question, not a generic textbook line.

1.3 Types of data and the large data set

Concept

Data is classified along two independent axes. **Qualitative** data records categories or labels (e.g. weather type: “rain”, “sun”); **quantitative** data records numerical quantities (e.g. temperature in °C). Quantitative data is further split into **discrete** (can only take specific, countable values — e.g. number of rainy days, which can only be a whole number) and **continuous** (can take any value in a range, limited only by the precision of measurement — e.g. rainfall in mm).

Data is also described by how it was gathered: **primary data** is collected first-hand by the person carrying out the analysis, for this specific study; **secondary data** was collected by someone else for a different original purpose (this includes almost all published data sets, such as the large data set described below).

The **large data set** referenced in 'Why this chapter matters' is a specific, named secondary data set (for Edexcel: daily weather data for several UK and overseas locations across selected months in one or more years) that the exam board publishes in advance and expects you to have explored during the course. Questions can ask you to identify the type of a specific variable within it, interpret an extract of it, or use it as the context for a distribution or hypothesis test later in the course — so it is worth exploring the data set itself early in the course, rather than leaving it as last-minute revision.

Procedure

When classifying a variable from a data set:

1. Ask whether it is a category/label or a number — if a label, it is qualitative; if a number, continue.
2. Ask whether it could take a value in between two consecutive possible values (e.g. between 3 and 4) — if yes, it is continuous; if not, and it can only jump between fixed values (e.g. whole numbers only), it is discrete.
3. Ask who collected it and why — if the person analysing it collected it themselves for this purpose, it's primary; if it was already collected for another purpose (including any published or provided data set), it's secondary.

Worked example

EXAMPLE 1.3

The table below is an extract from a large weather data set.

Location	Daily mean temperature (°C)	Total daily rainfall (mm)	Weather type
Leeming	12.4	0.0	Sunny
Camborne	14.1	3.2	Rain

(a) State, with a reason, whether “Daily mean temperature” is discrete or continuous. (b) State, with a reason, whether “Weather type” is qualitative or quantitative. (c) State, giving a reason, whether this is primary or secondary data for a student using it to answer exam questions.

Step 1 (part a). Continuous — temperature can take any value within a range (e.g. 12.4°C is only limited by the precision of the thermometer, not by fixed allowed values). (B1)

Step 2 (part b). Qualitative — “Sunny” and “Rain” are categories/labels, not numerical quantities. (B1)

Step 3 (part c). Secondary — the data was originally collected by a weather station/meteorological service for its own purposes, not by the student. (B1)

Answer (a) Continuous. (b) Qualitative. (c) Secondary.

EXAM TIP

It is tempting to call temperature “discrete” because it's given to 1 decimal place in the table — but rounding for display doesn't change the underlying variable's type; the test is whether the *true* quantity could take any in-between value, not how it happens to be recorded.

Chapter summary

Idea	Key point
Census vs sample	A census surveys the whole population; almost always impractical, so a sample is used instead
Sampling frame	The list a sample is drawn from — must match the population or bias results
Simple random sampling	Every member equally likely to be chosen; needs a full sampling frame
Systematic sampling	Fixed interval k from a random start; watch for hidden periodicity
Stratified sampling	Proportional allocation by group: stratum size $\times$ (sample size $\div$ population size)

Idea	Key point
Quota / opportunity sampling	Fast and cheap but non-random, so prone to bias
Qualitative vs quantitative	Category/label vs numerical value
Discrete vs continuous	Fixed, countable values vs any value in a range
Primary vs secondary data	Collected by the analyst themselves vs collected by someone else, including the large data set

Common exam mistakes

- Saying a sample “might be inaccurate” instead of naming which specific group is over- or under-represented and why.
- Forgetting to round stratified sampling answers to whole numbers, or being alarmed when rounded values don't sum exactly to the target sample size (this is expected and fine).
- Classifying data type by how it's *displayed* (e.g. rounded to 1 d.p.) rather than by what the underlying variable can actually take.
- Confusing a sampling frame with the sample itself — the frame is the *list you could choose from*, not the people actually chosen.

You should now be able to

- Explain when a census is preferable to a sample, and vice versa.
- Identify and evaluate the sampling frame for a given scenario, including sources of bias.
- Describe simple random, systematic, stratified, quota and opportunity sampling, and evaluate each for a given context.
- Carry out a stratified sampling calculation, including proportional allocation and rounding.
- Classify a variable as qualitative/quantitative, discrete/continuous, and primary/secondary, including variables from the large data set.